

Employment

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| Graduate Research Assistant                                                                                                                                                                                                                                                                                                                                                                      | NC State University  | Apr 2025 – Present  |
| <ul style="list-style-type: none"><li>Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets.</li><li>Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.</li></ul>         |                      |                     |
| Product Engineer Intern                                                                                                                                                                                                                                                                                                                                                                          | Isomer AI            | May 2025 – May 2026 |
| <ul style="list-style-type: none"><li>Collaborated with product and design teams to ship 10+ production AI features: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows; 923 commits, 7 production deployments.</li><li>Owned full-stack React/Node.js + TypeScript codebase: PostgreSQL, Redis, Docker, Auth0 RBAC, and Playwright test coverage.</li></ul>      |                      |                     |
| Software Developer Intern                                                                                                                                                                                                                                                                                                                                                                        | Mphasis (HP Company) | Jun 2024 – Aug 2024 |
| <ul style="list-style-type: none"><li>Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims.</li><li>Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.</li></ul> |                      |                     |
| AI/ML Intern                                                                                                                                                                                                                                                                                                                                                                                     | Capgemini            | Jun 2023 – Aug 2023 |
| <ul style="list-style-type: none"><li>Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages.</li><li>Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load.</li></ul>             |                      |                     |

Education

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| Raleigh, NC                                                                                                                                                                                                                                                                                    | North Carolina State University | Aug 2024 – May 2026 |
| <ul style="list-style-type: none"><li>Master of Science in Computer Science</li><li>Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.</li></ul> |                                 |                     |
| Mumbai, India                                                                                                                                                                                                                                                                                  | Mumbai University               | Aug 2020 – May 2024 |
| <ul style="list-style-type: none"><li>Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS &amp; Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.</li></ul>                                         |                                 |                     |

Projects

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| PulseRank                                                                                                                                                                                                                                                                                                                                                                        | Real-Time Leaderboard & Activity Feed | 2026 |
| <ul style="list-style-type: none"><li>Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js; MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; live SSE.</li><li>Validated multi-DB consistency with Testcontainers; benchmarked hot paths with k6 at 10K concurrent users.</li></ul> |                                       |      |
| CortexQ                                                                                                                                                                                                                                                                                                                                                                          | Kubernetes LLM Inference Platform     | 2026 |
| <ul style="list-style-type: none"><li>Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.</li><li>Deployed via Helm + ArgoCD GitOps on Kubernetes; load tested k6 to 500 VUs; 88 pytest integration tests.</li></ul>         |                                       |      |
| LLM Eval Harness                                                                                                                                                                                                                                                                                                                                                                 | Independent Project                   | 2025 |
| <ul style="list-style-type: none"><li>Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, reasoning, and code generation tasks.</li><li>Tracked hallucination rate, p50/p95 latency, and cost-per-call; applied McNemar significance tests across 6 metrics.</li></ul>      |                                       |      |

Languages and Technologies

- Languages:** TypeScript, JavaScript, Java, Python, SQL, Bash
- Frontend & Full Stack:** React, Next.js, Node.js, NestJS, REST APIs, PostgreSQL, Redis, MongoDB, MySQL, Auth0
- Testing & CI/CD:** Playwright, Testcontainers, k6, pytest, GitHub Actions, Docker, Kubernetes, Helm, ArgoCD
- Cloud & Observability:** AWS, GCP, OpenTelemetry, Prometheus, Grafana, Sentry, Kafka, LangChain

Additional Experience and Awards

- IEEE Published:** Co-authored and presented research on Computer Vision-Based Cybersecurity Threat Detection at the Springer icSoftComp 2023 international conference, reaching an audience of 500+ global attendees.
- Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; built 3 full-stack MVPs end-to-end in 48-hour competitive sprints.
- GDSC Lead:** Led 4+ tech events as Google Developer Student Club Lead, reaching 500+ students across hackathons and conferences.
- Technical Writing:** Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog.